

Purdue University

Grid Connected Single Phase Double Stage Inverter Control with PV Array, Boost Converter and MPPT Control

Project # 2

Majharul Hoque
11-10-2024

ECE 61016: Power Electronic Converters and Systems

Purdue University

Project Assignment #2

Name: Majharul Hoque

1. Design a solar panel based on the specification given below:

Solar panel rating (single array)

$$V_{mpp} = 51.671 \text{ V}$$

$$I_{mpp} = 6.06 \text{ A}$$

$$P_{mpp} = 313.2 \text{ W}$$

There are 2 arrays parallel connected, and 3 series connected providing $6 \times P_{mpp}$ when the irradiance is at $1,000 \text{ W/m}^2$.

Electrical Characteristics of a typical Maxeon Gen III Cell At Standard Test Conditions (STC) STC: 1000 W/m^2 , AM 1.5G and cell temp 25°C							
	Cell Bin	Pmpp (Wp)	Eff. (%)	Vmpp (V)	Impp (A)	Voc (V)	Isc (A)
Max Peak Performance	Me3	3.89	25.1	0.640	6.08	0.727	6.45
Max Premium Performance	Le3	3.84	24.8	0.634	6.06	0.724	6.43
Max High Performance	Ke3	3.77	24.4	0.627	6.02	0.721	6.41

Electrical parameters are nominal values.

Temp.Coefficients in SunPower Panels: Voltage: $-1.74 \text{ mV}/^\circ\text{C}$, Current: $2.9 \text{ mA}/^\circ\text{C}$,
Power: $-0.29\%/^\circ\text{C}$

- a) Use MATLAB Simulink to model the above solar panel and plot the IV curve when the irradiances are 1000, 800, 600, 400, and 200 W/m^2 .
- b) Design a boost converter has an input voltage of PV V_{mpp} and the required output voltage is 300 V for single-phase grid connection. The switching frequency is at 7 kHz. Design LC filter that meets:
 - a. Inductor ripple current $< 15\%$ of the output current
 - b. Output capacitor value $> 5000 \mu\text{F}$
- c) Implement MPPT controller block with perturb and observe algorithm and validate the operation with different irradiances (1000 and 500 W/m^2)

A general subsystem of the PV based Boost converter and use of MPPT system to keep output at the maximum level is shown below:

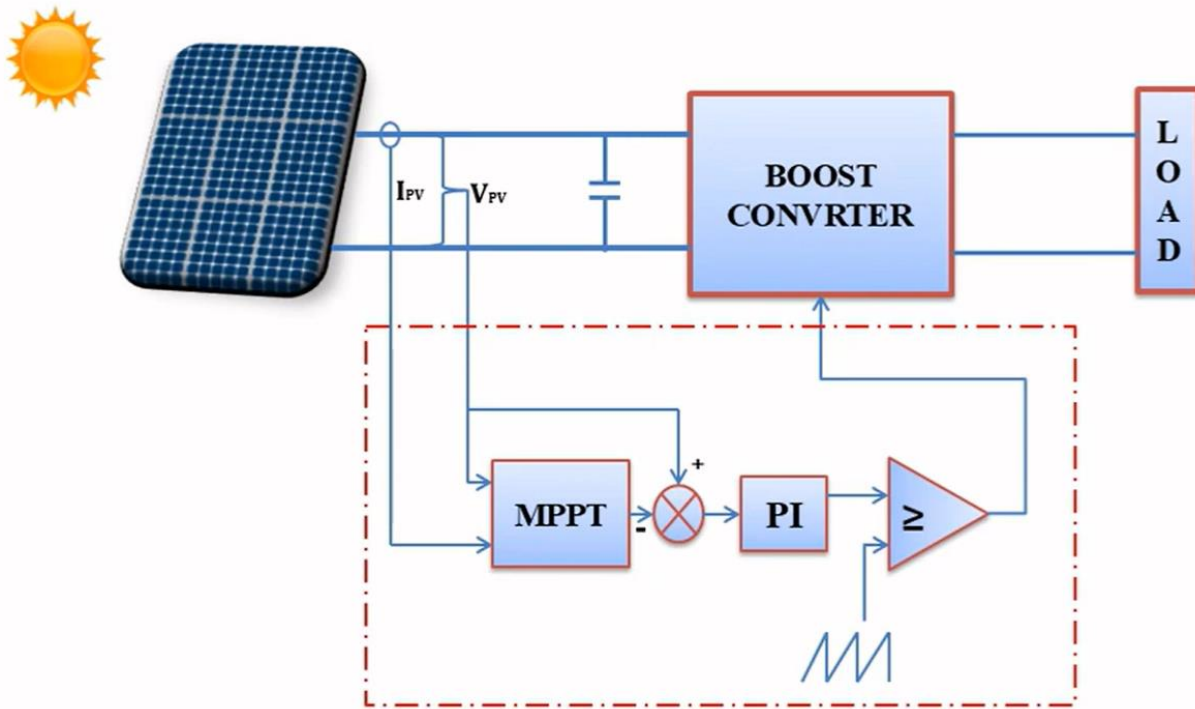


Figure 1: A general PV based Boost converter with MPPT algorithm implementation for Maximum Power Point.

A SIMULINK model of the specified solar panel or PV system has been shown below:

- The design is a general solar PV system. No commercially available PV system is used for the simulation:

Solar panel rating (single array)

$V_{mpp} = 51.671 \text{ V}$

$I_{mpp} = 6.06 \text{ A}$

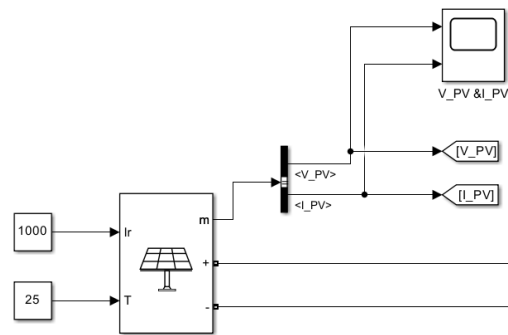
$P_{mpp} = 313.2 \text{ W}$

Electrical Characteristics of a typical Maxeon Gen III Cell							
At Standard Test Conditions (STC)							
STC: 1000W/m ² , AM 1.5G and cell temp 25°C							
	Cell Bin	P _{mpp} (Wp)	Eff. (%)	V _{mpp} (V)	I _{mpp} (A)	V _{oc} (V)	I _{sc} (A)
Max Peak Performance	Me3	3.89	25.1	0.640	6.08	0.727	6.45
Max Premium Performance	Le3	3.84	24.8	0.634	6.06	0.724	6.43
Max High Performance	Ke3	3.77	24.4	0.627	6.02	0.721	6.41

Electrical parameters are nominal values.

Temp.Coefficients in SunPower Panels: Voltage: -1.74mV/°C, Current: 2.9mA/°C,
Power: -0.29%/°C

The below SIMULINK model is only for the PV system. The settings have been shown below :



Input 1 = Sun irradiance, in W/m2, and input 2 = Cell temperature, in deg.C.

Parameters		Advanced	
Array data			
Parallel strings	2		
Series-connected modules per string	3		
Module data			
Module:	User-defined		
Maximum Power (W)	314.1504		
Cells per module (Ncell)	81		
Open circuit voltage Voc (V)	81*0.727	58.887	
Short-circuit current Isc (A)	6.45		
Voltage at maximum power point Vmp (V)	0.640*81	51.84	
Current at maximum power point Imp (A)	6.06		
Temperature coefficient of Voc (%/deg.C)	-1.74e-3/0.727*100	-0.23934	
Temperature coefficient of Isc (%/deg.C)	2.9e-3/6.45*100	0.044961	
Model parameters			
Light-generated current IL (A)	6.4513		
Diode saturation current IO (A)	1.3343e-12		
Diode ideality factor	0.96977		
Shunt resistance Rsh (ohms)	326.9714		
Series resistance Rs (ohms)	0.064206		

Calculations for the Combined Array

1. Cells per module (Ncell):

Each panel typically has 81 cells.

2. Open Circuit Voltage (Voc):

The open-circuit voltage for one panel is 51.671 V. Since there are 3 panels in series, the total open-circuit voltage is: $V_{oc_total} = 3 \times 51.671 = 155.013 \text{ V}$.

3. Short-Circuit Current (Isc):

The short-circuit current for one panel is 6.06 A. With two parallel arrays, the total short-circuit current is: $I_{sc_total} = 2 \times 6.06 = 12.12 \text{ A}$.

4. Voltage at Maximum Power Point (Vmp):

The voltage at the maximum power point for one panel is 51.671 V. With three panels in series, the total maximum power point voltage is: $V_{mp_total} = 3 \times 51.671 = 155.013 \text{ V}$

5. Current at Maximum Power Point (Imp):

The current at the maximum power point for one panel is 6.06 A. With two parallel arrays, the total current at maximum power is: $I_{mp_total} = 2 \times 6.06 = 12.12 \text{ A}$

6. Maximum Power (W):

The maximum power output for one panel is 313.2 W. With 6 panels (2 in parallel and 3 in series), the total maximum power is: $P_{total} = 6 \times 313.2 = 1879.2 \text{ W}$

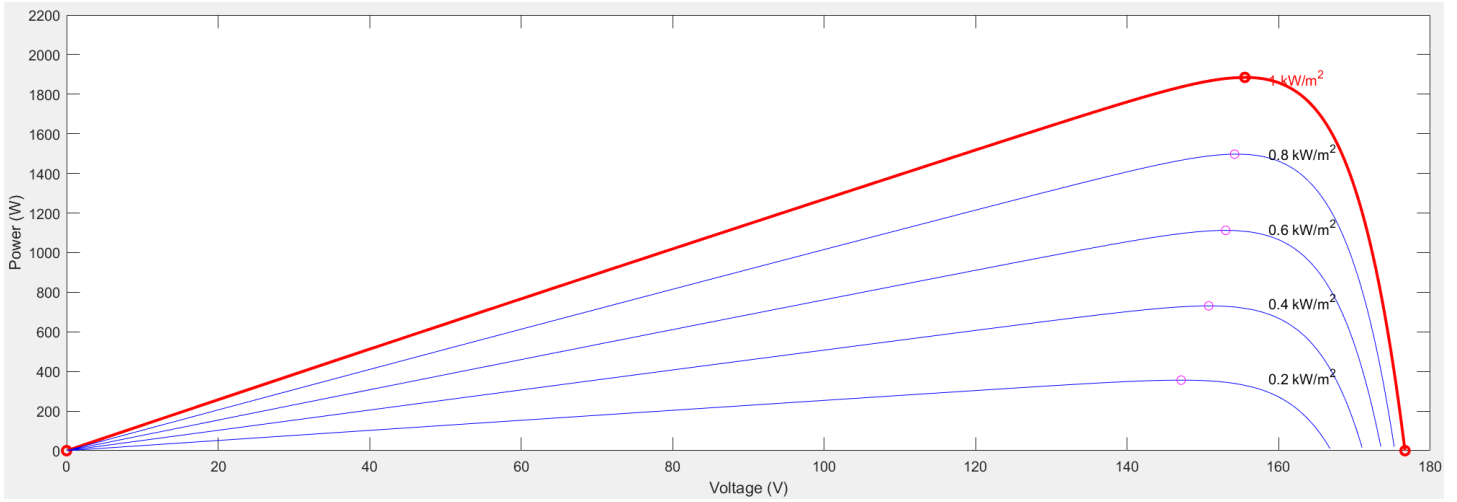


Figure 2: Voltage Vs. Power plot for the specified settings.

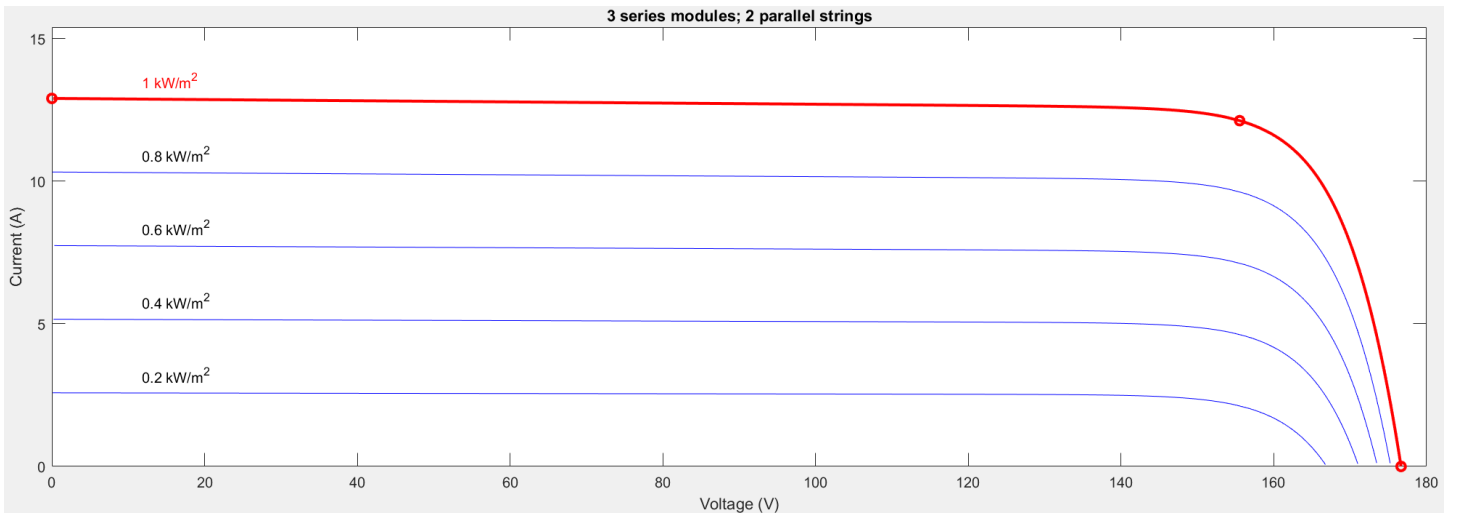


Figure 3: Voltage Vs. Power plot for the specified settings.

Now, the PV system has been connected to a Boose converter as shown in the model below. Here:

Input Voltage, $V_{in} = 3 \times V_{mpp} = 155.013 \text{ V}$

Output Voltage, $V_{out} = 300 \text{ V}$

Switching Frequency, $f_{sw} = 7 \text{ kHz}$

Duty cycle, $D = 1 - \frac{V_{in}}{V_{out}} = 0.483$

Inductor ripple current, $\Delta I_L < 15\% \text{ of } I_{out} = 0.15 \times 6.264 \text{ A} = 0.9396 \text{ A}$

Capacitor, $C < 5000 \text{ } \mu\text{F}$

Inductor, $L = \frac{V_{in} \times D}{f_{sw} \times \Delta I_L} = \sim 12 \text{ mH}$

A 5000Ω load resistor is placed at the output.

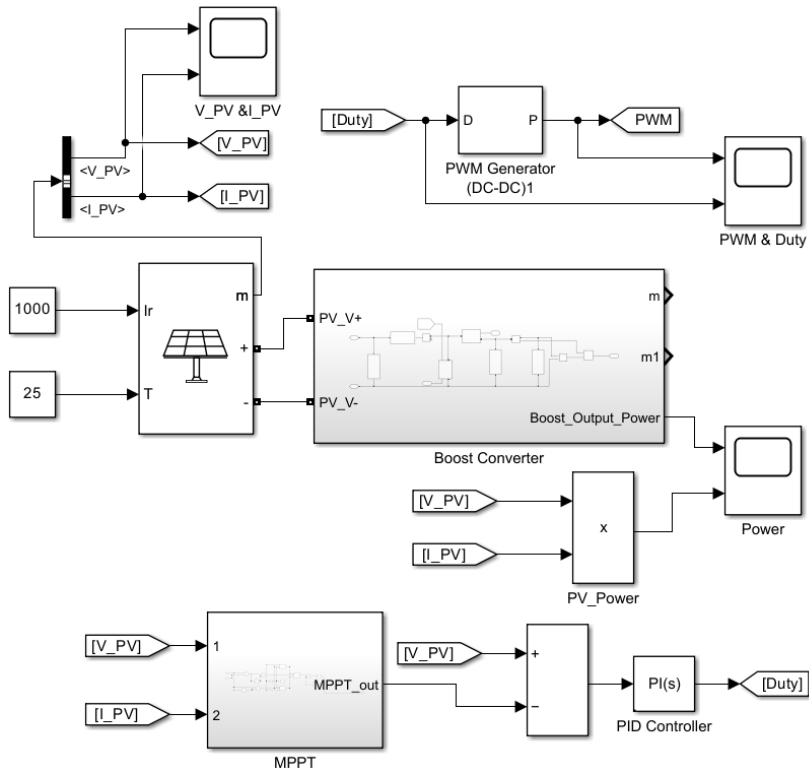


Figure 4: The PV system is connected to a Boost Converter with a MPPT controller block is also shown.

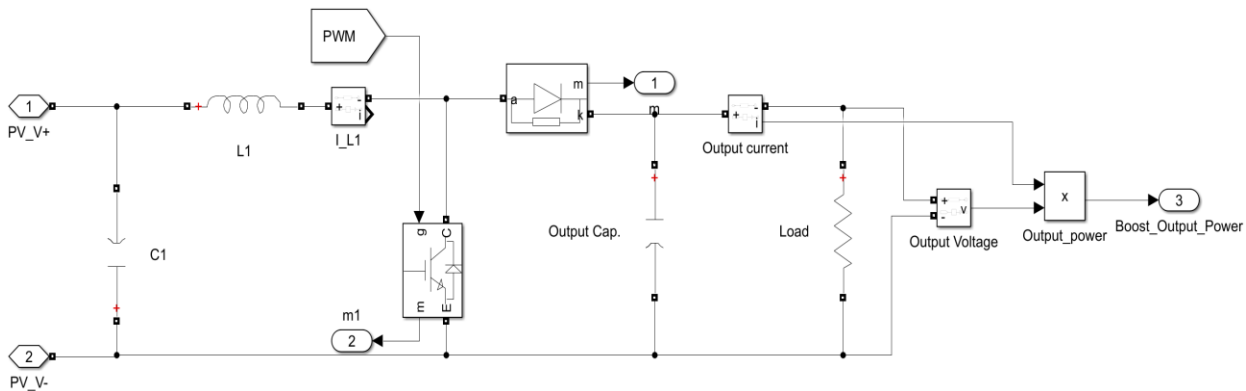


Figure 5: The Boost converter design in SIMULINK.

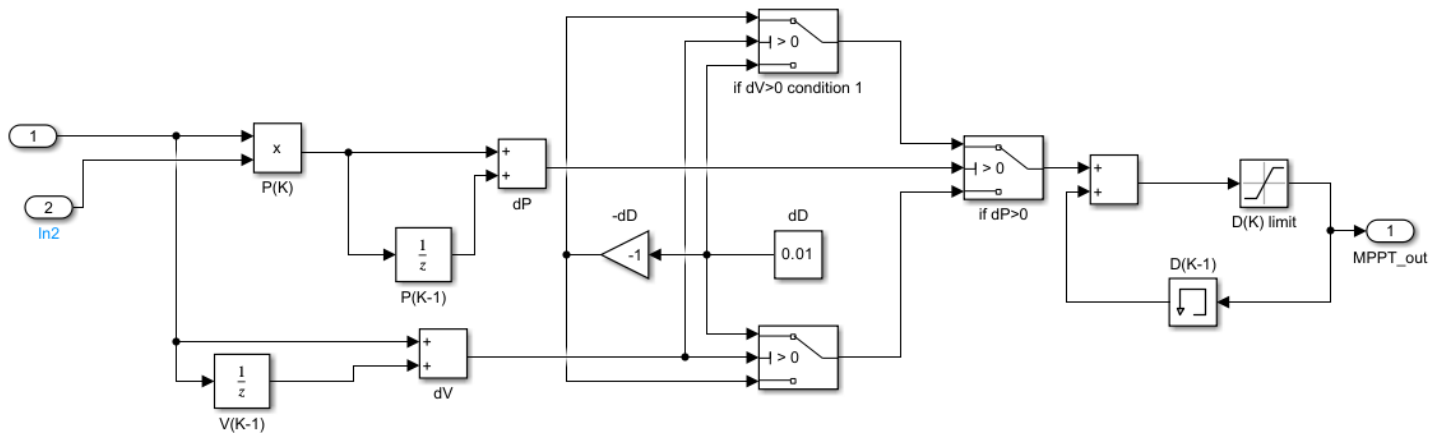


Figure 6: Maximum Power Point Tracking Controller block.

MPPT Controller Using Perturb and Observe:

The P&O algorithm adjusts the voltage to find the maximum power point (MPP). If increasing voltage results in increased power, the controller continues in that direction. If power decreases, it reverses the direction.

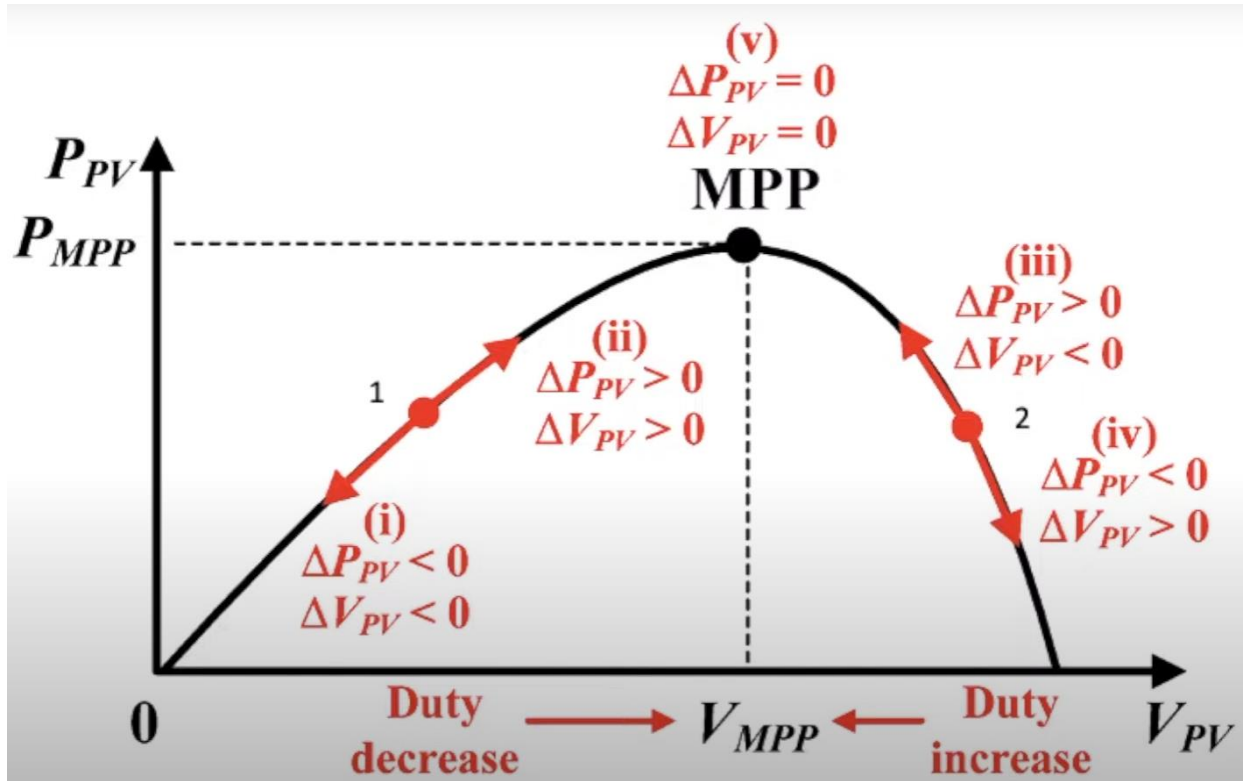


Figure 7: MPPT subsystem tracks the Output power and keeps tries to keep it at the maximum level.

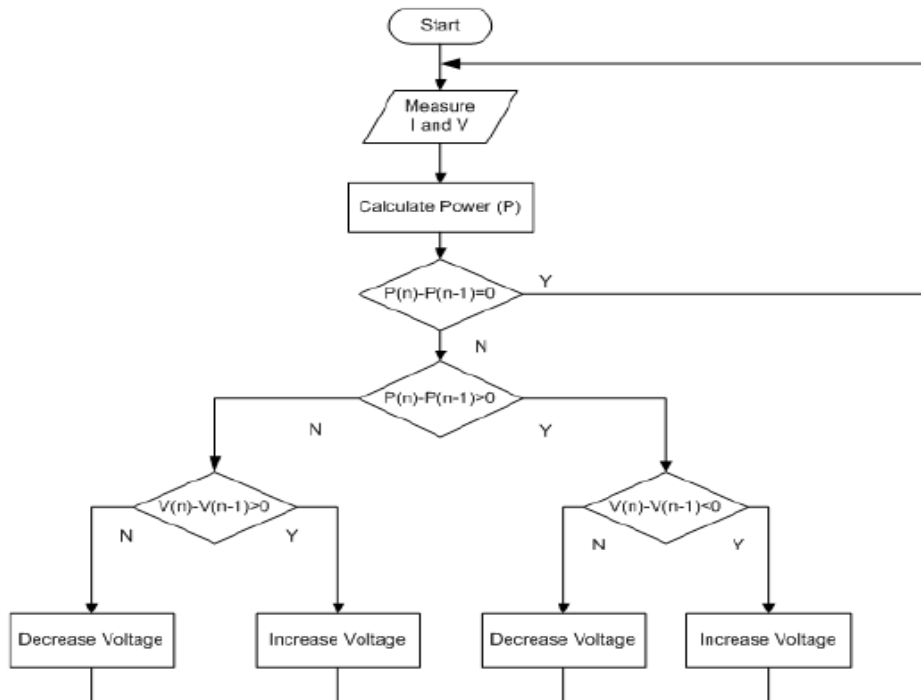


Figure 8: Flowchart of the MPPT algorithm.

In the below images for difference irradiance levels the MPPT keeps the system tracking the maximum power point level. The first sets of Power, Voltage and current plots are for 1000 W/m2 irradiance level.

Irradiance level: 1000 W/m2

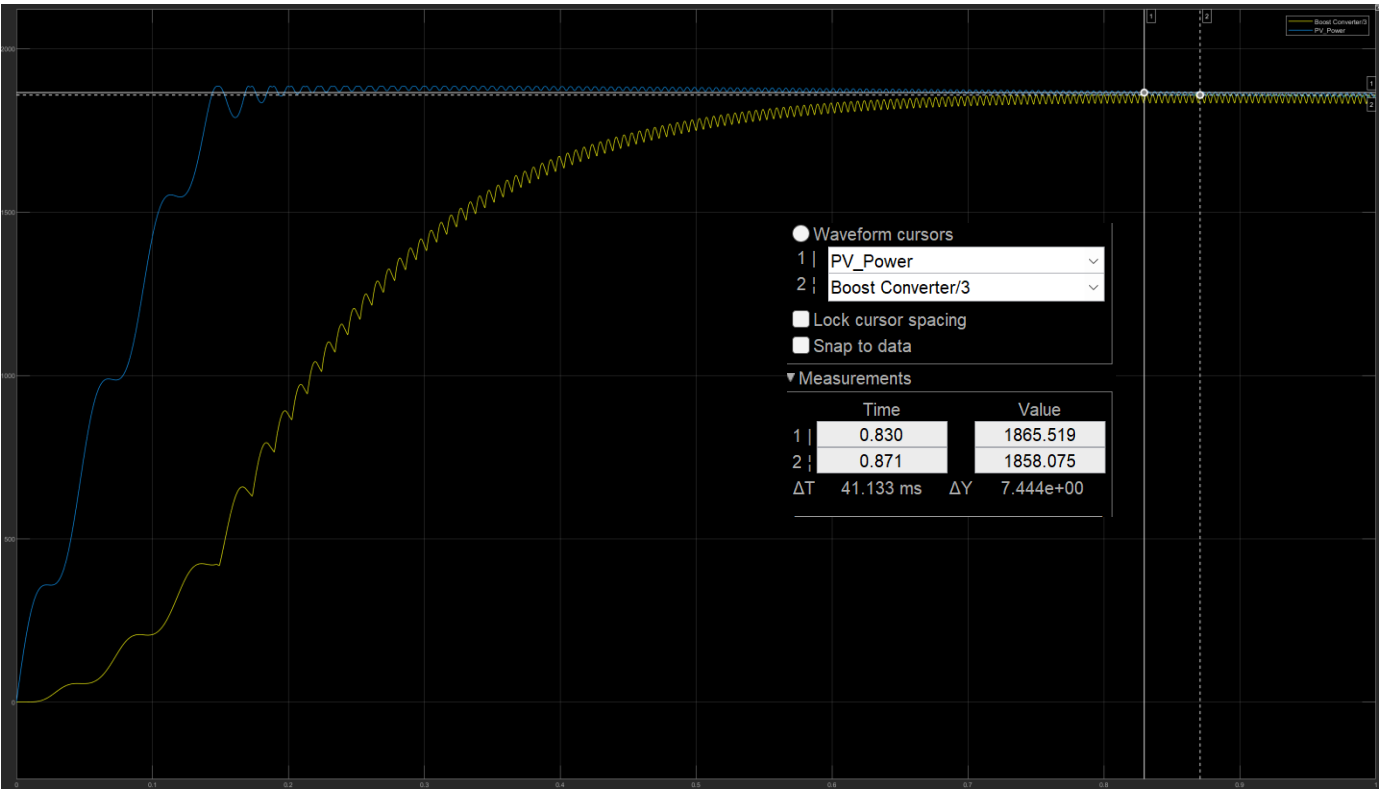


Figure 9: PV Output power Vs. Boost Converter Output power.

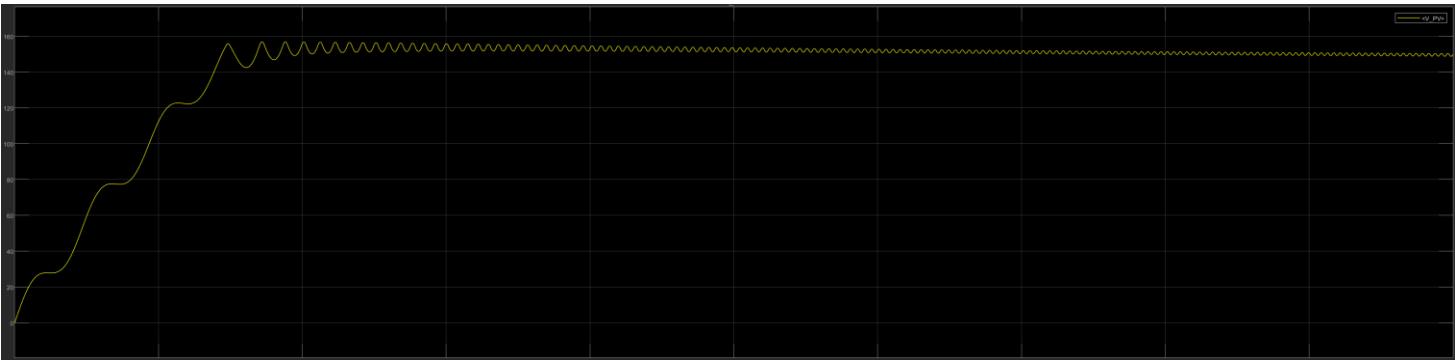


Figure 10: MPPT tracking in work, keeping the voltage around ~155V level

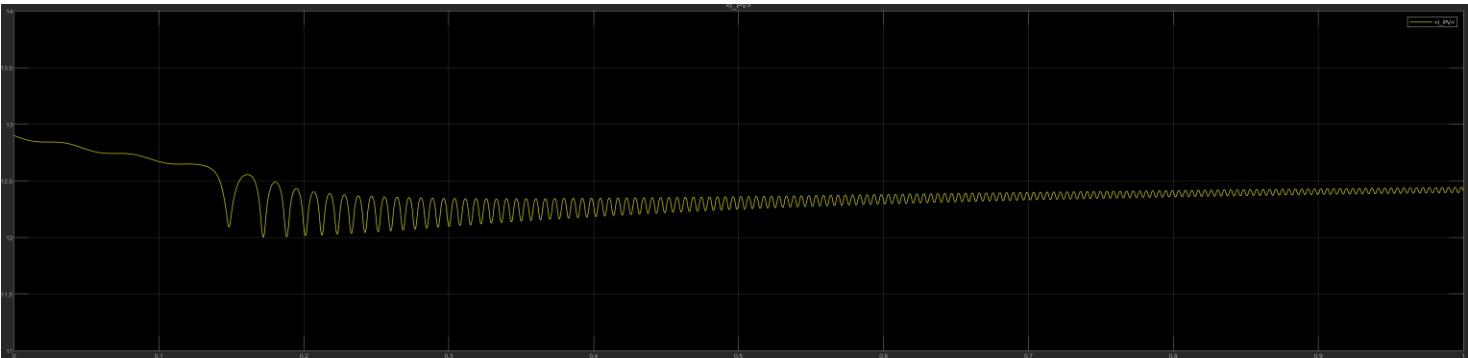


Figure 11: Output Current of the PV system around ~12.5V

Irradiance level: 500 W/m2

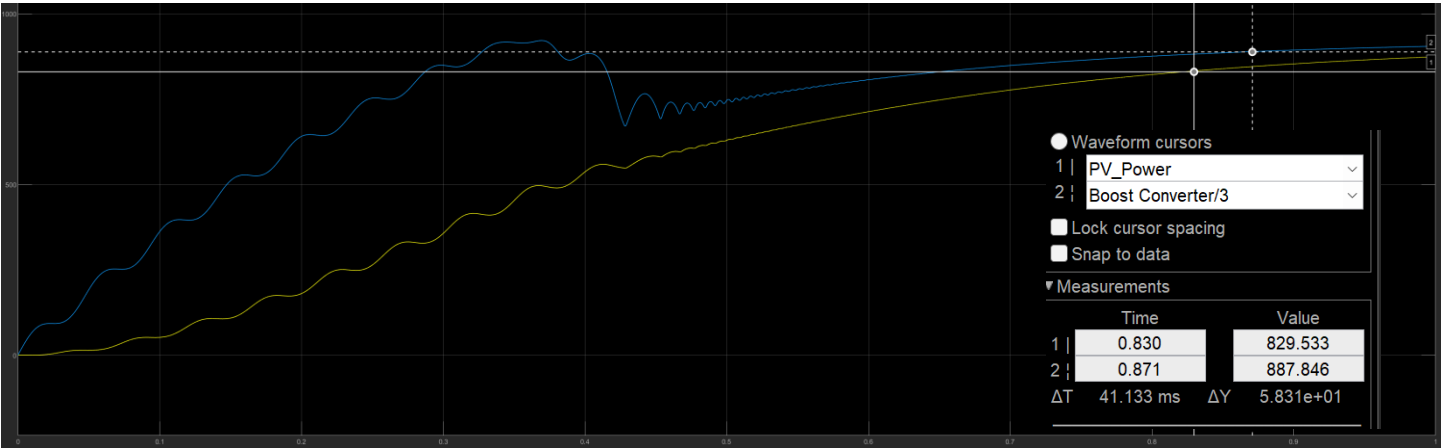


Figure 12: AT 500 W/m2 irradiance level the output power went down but still the MPPT is keeping the system at Maximum Power Level.

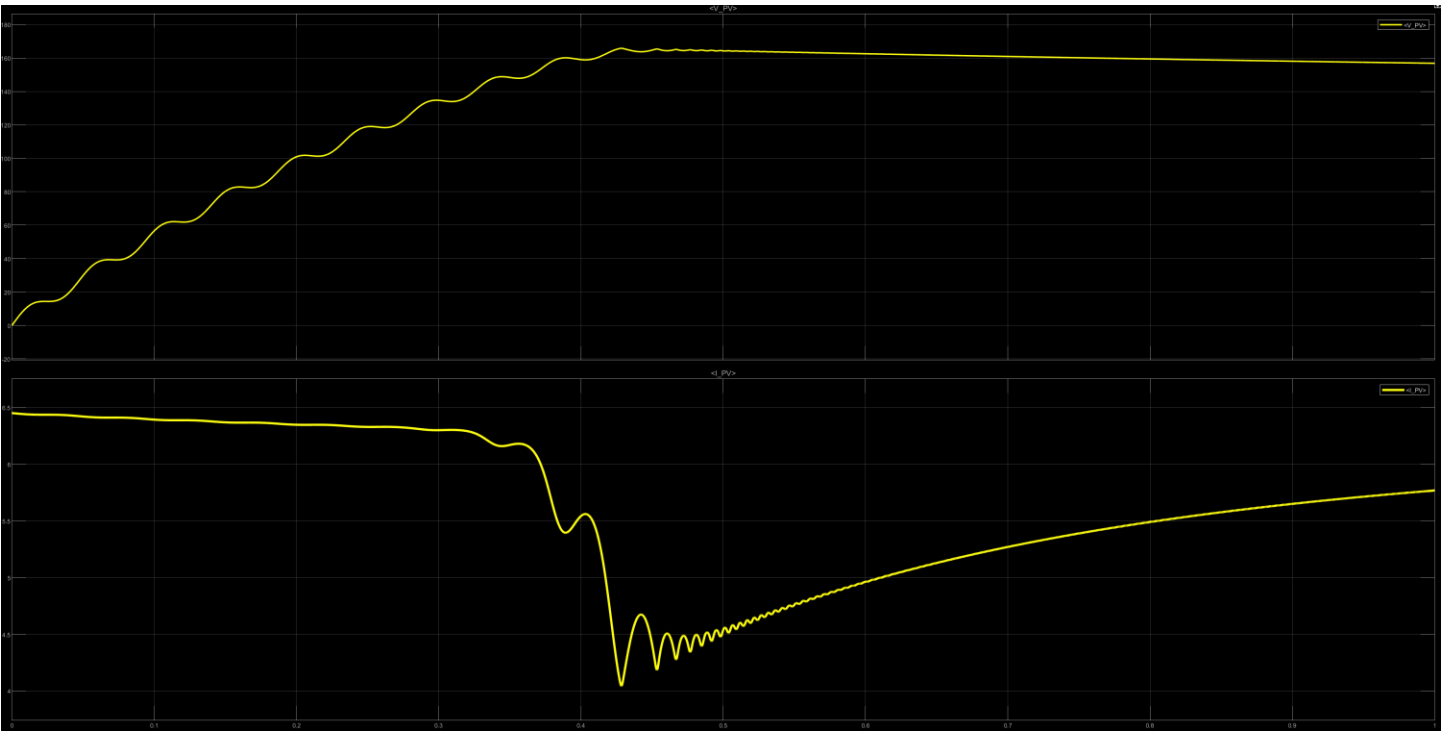


Figure 13: Output Voltage around ~160V and Current of the PV system around ~12.5A

2. Design a single-phase two-level inverter using MATLAB Simulink with below spec:

DC-link voltage: 300 V

Fundamental frequency: 60 Hz

Switching frequency: 10 kHz

Sampling frequency: 2 MHz

Output load: 5 ohm and 5 mH (use single-phase series RLC branch in element tab)

- Plot 0.017s of fundamental signal and triangular waveforms.
- Record and plot 0.1s of line-to-line, line-to-neutral, and phase current waveform at modulation index 1.
- Record and plot 0.1s of line-to-line, line-to-neutral, and phase current waveform at modulation index 0.5.

See the SIMULINK design of a single-phase two-level inverter below:

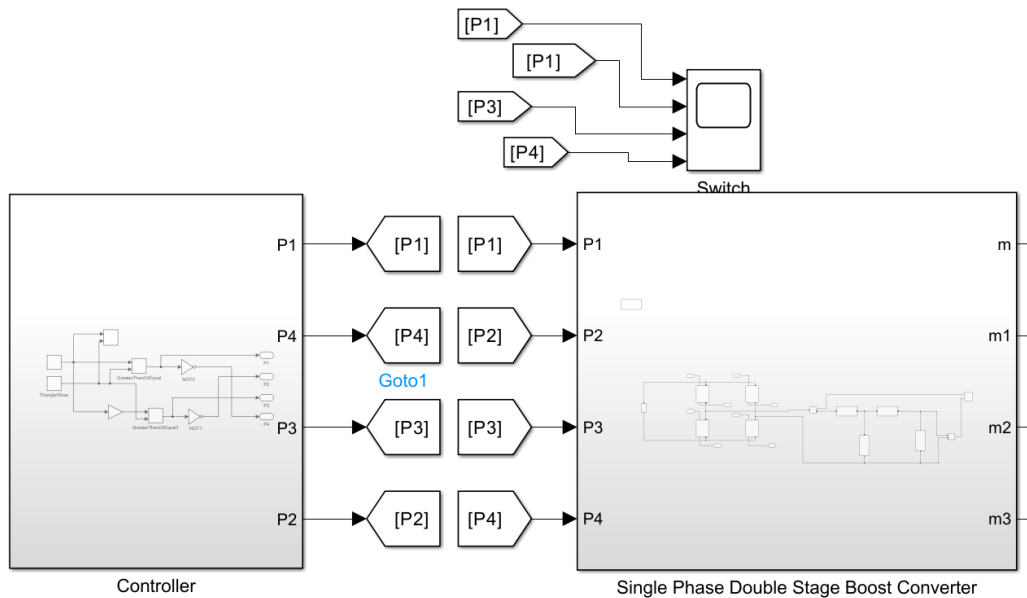


Figure 14: single-phase two-level inverter using MATLAB Simulink

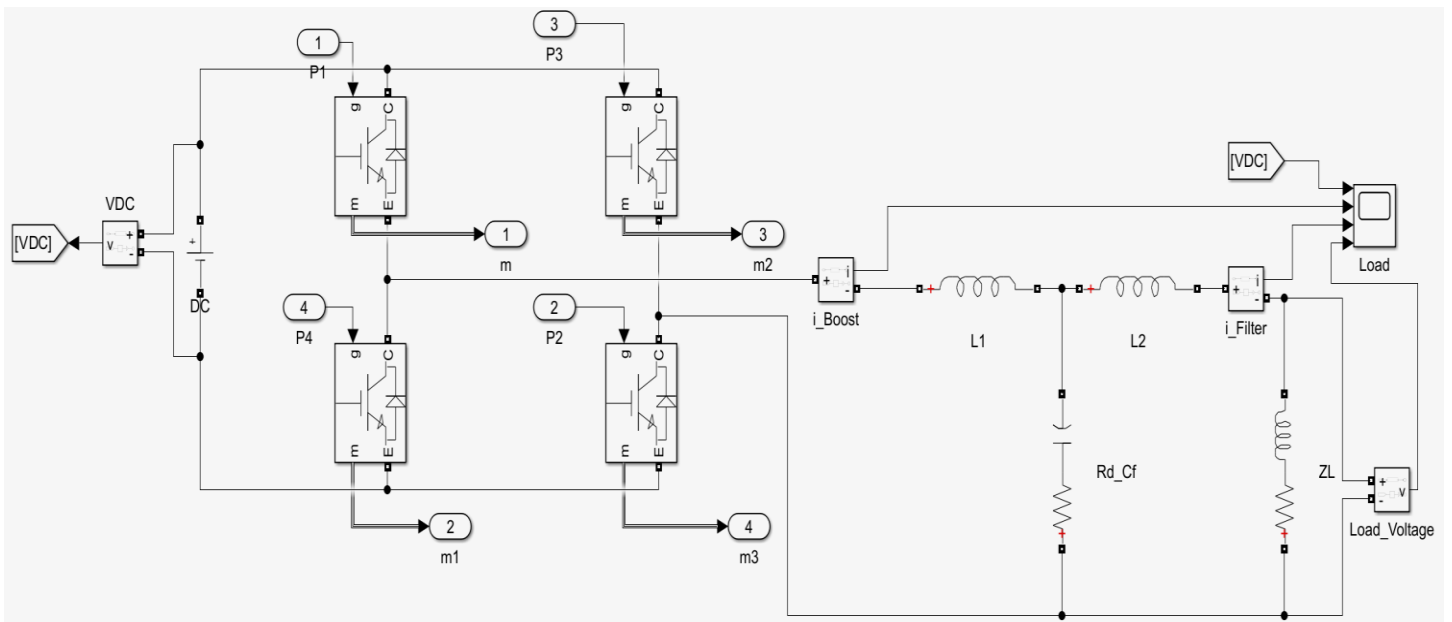


Figure 15: The Inverter Circuit with VDC = 300V

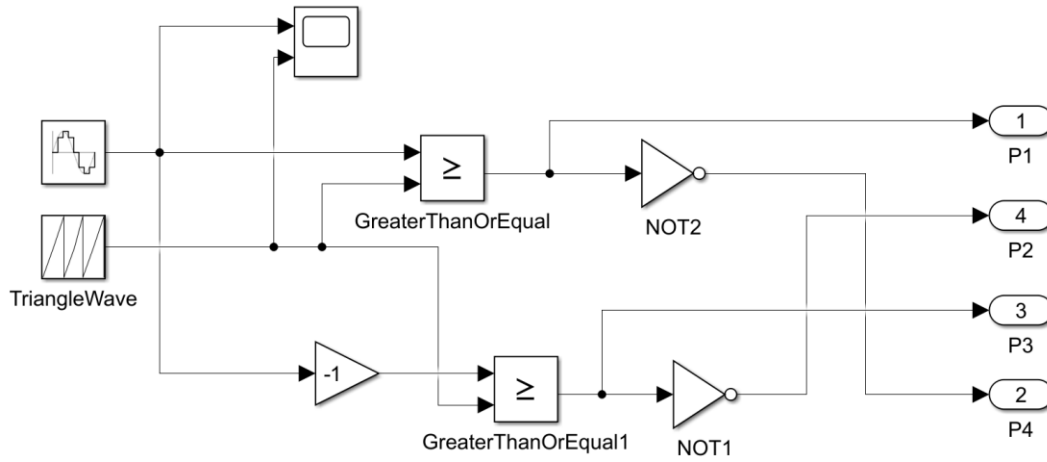


Figure 16: Controller switch logic for the Double stage IGBT switches.

Parameters for Repeating Sequence (Triangular Carrier Wave)

1. Time Values:

Since the switching frequency is 10 kHz, the period T of the triangular wave is:

$$T = \frac{1}{10\text{kHz}} = 0.0001 \text{ s (or } 100 \mu\text{s)}$$

The triangular wave will go from 0 to its peak and back to 0 within this period. To define the time values, set it to half of this period for rising and falling slopes:

$$\text{Time values} = [0, \frac{T}{2}, T] = [0, 50 \times 10^{-6}, 100 \times 10^{-6}]$$

2. Output Values:

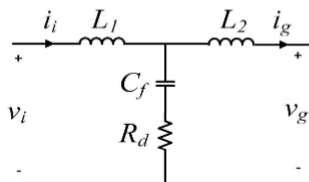
The carrier wave's peak should match the DC-link voltage for modulation purposes. Since the DC-link voltage is 300V, the triangular wave can vary between 0 and 300 V. Therefore:

$$\text{Output values} = [0, 300, 0]$$

The LCL filter seen at the output of the inverter has been chosen because it has better characteristics compared to L and LC filter. Since the Inverter is designed for 10kHz switching frequency, so the values of Capacitor and inductors suggested in the lecture note has been used.

Output LCL Filter Design

- LCL filter has better attenuation with damping and size is smaller than L and LC filter
- Dynamic frequency response of grid current



Per phase LCL filter model with passive damping resistance

i_i : inverter output current
 v_i : phase voltage of inverter output
 L_1 : the inverter-side inductance
 L_2 : the grid-side inductance
 C_f : the paralleled filter capacitor
 R_d : the damping resistance
 v_g : phase voltage of grid
 i_g : grid current

- Output LCL filter designed as switching frequency increases
- C_f is set to be 5.6 μF for limitation of 5% reactive power from capacitor

Parameter	Switching Frequency			
	10 kHz	20 kHz	50 kHz	100 kHz
L_1	1 mH	600 μH	300 μH	150 μH
C_f	5.6 μF	5.6 μF	5.6 μF	5.6 μF
L_2	250 μH	65 μH	10 μH	10 μH
R_d	5.97 Ω	3.2 Ω	1.3 Ω	1.27 Ω
f_{res}	4.75 kHz	8.78 kHz	21.6 kHz	21.9 kHz

Filter size significantly reduced at high switching frequency

Figure 17: The LCL filter design at the output of the inverter.

The Fundamental frequency (60Hz) and the triangular waveform have been shown below. Comparing these two waveforms the gate voltages or the switching signals of the IGBT network is produced.

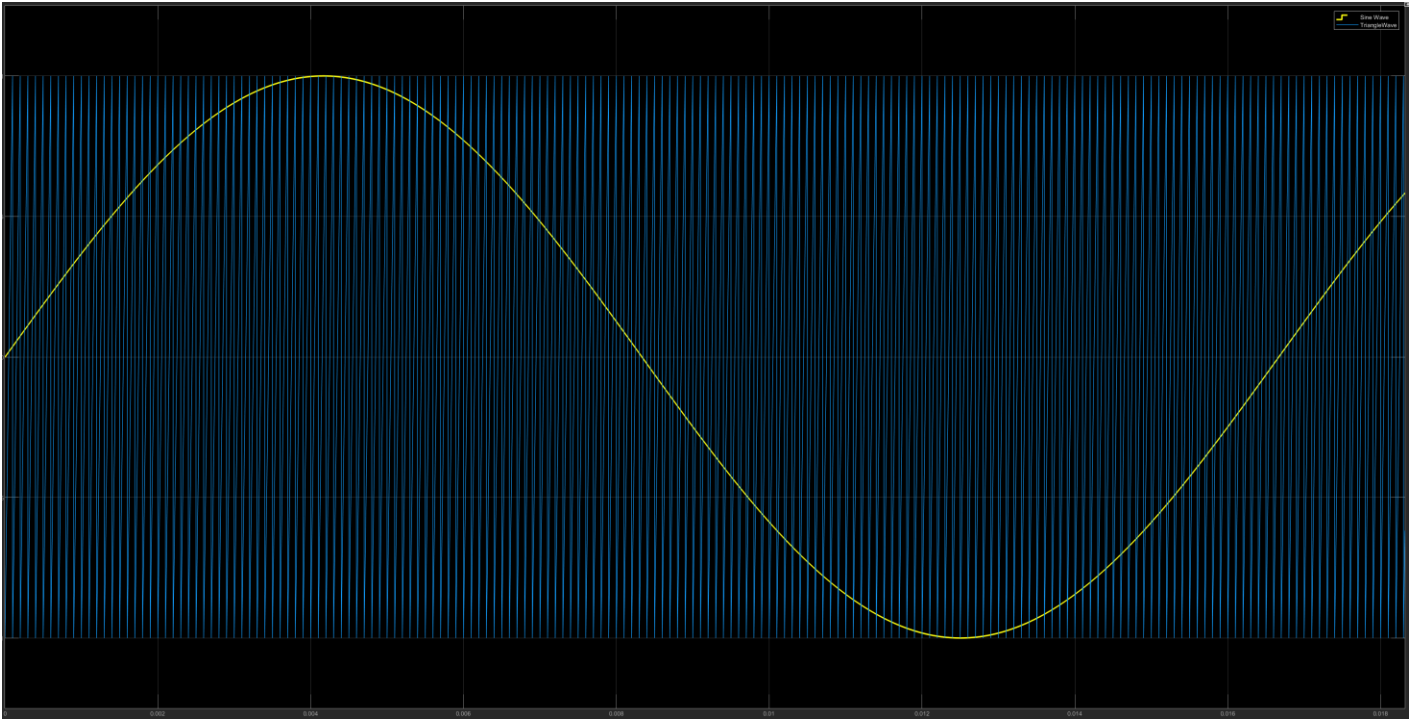


Figure 18: 0.018s of the fundamental signal and triangular waveform.

The Output voltage and current waveforms are also shown below when the modulation indexes are 1 and 0.5 respectively.



Figure 19: The yellow curve is the 300V DC voltage, The green signal is the periodic waveform at the Output of the inverter with ~150V peak, and the red curve is the in phase current waveform. The modulation index is set to 1.

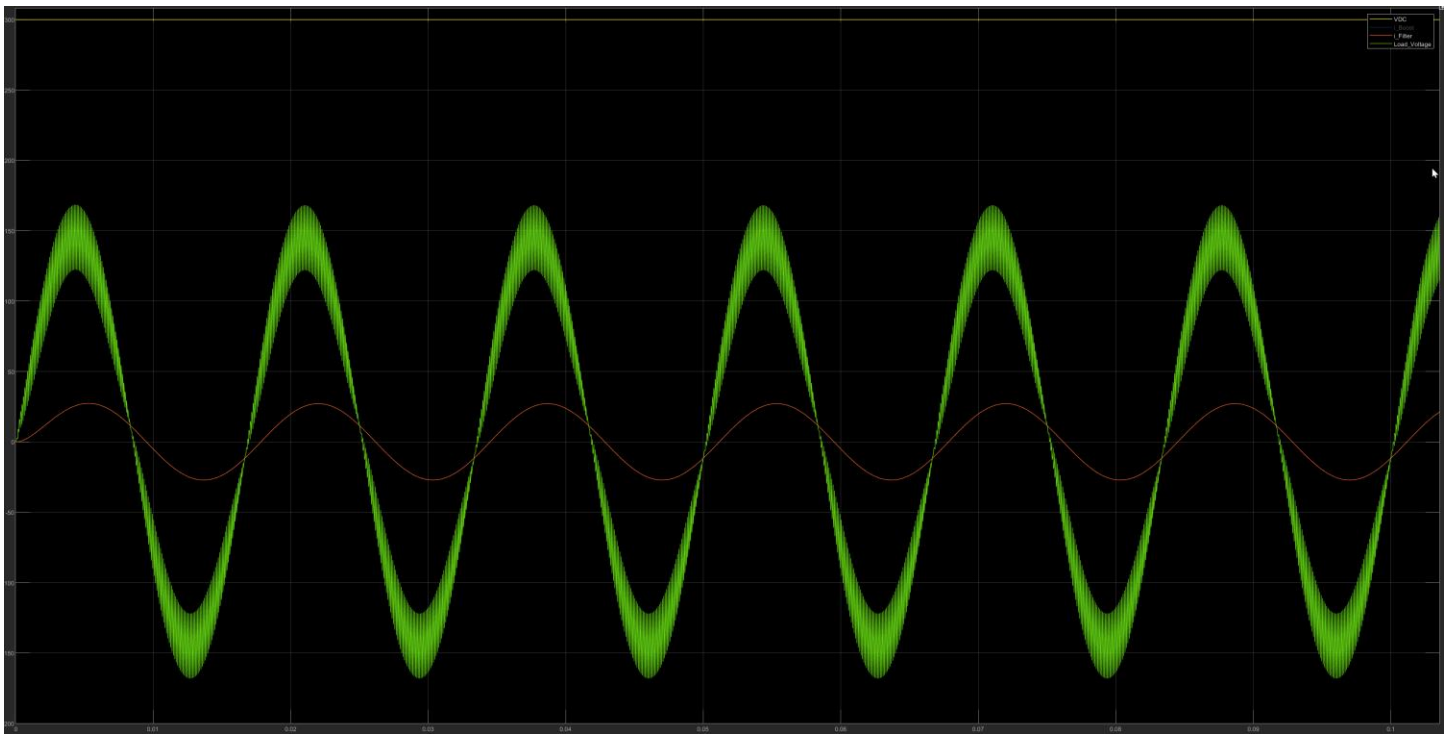


Figure 20: The yellow curve is the 300V DC voltage, The green signal is the periodic waveform at the Output of the inverter with ~150V peak, and the red curve is the in phase current waveform. The modulation index is set to 0.5.

Comparing Figures 19 and 20, the Output voltage is getting smoother for higher modulation index. But, as can be seen from Figure 19 the boost converter produced a periodic Sine waveform from a 300V DC voltage. The capacitors and inductors could be tuned better to improve smothering of the signals.

3. Connect PV, boost converter, and single-phase inverter and produce:
 - a. 1 kW active power to grid at 800 W/m^2 irradiance
 - b. 500 W active power with 500 VAR reactive power at 500 W/m^2 irradiance
 - c. 1 kVAR reactive power at 500 W/m^2 irradiance

The topology of the whole Double stage single phase Grid connected inverter with PV, Boost converter, MMPT control and PLL Vector control has been shown below:

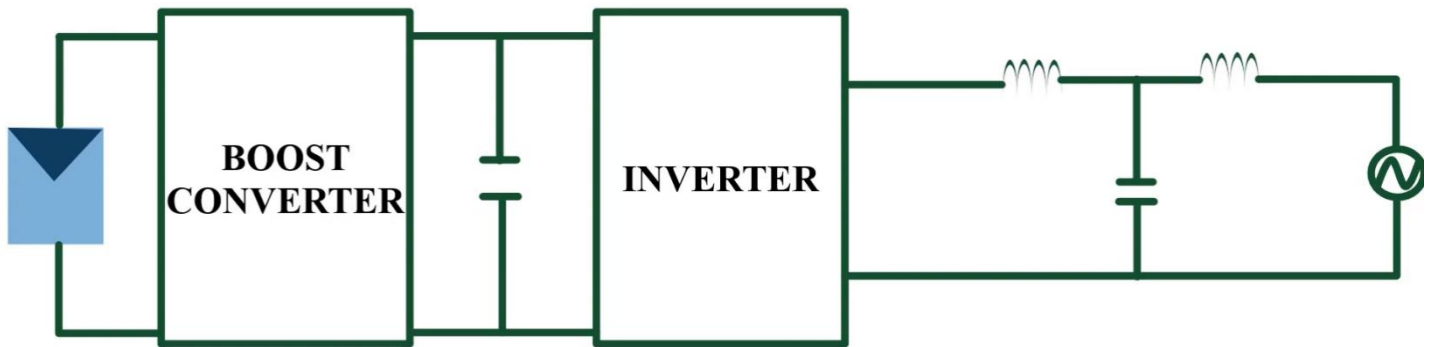


Figure 21: Simplified diagram of the whole system.

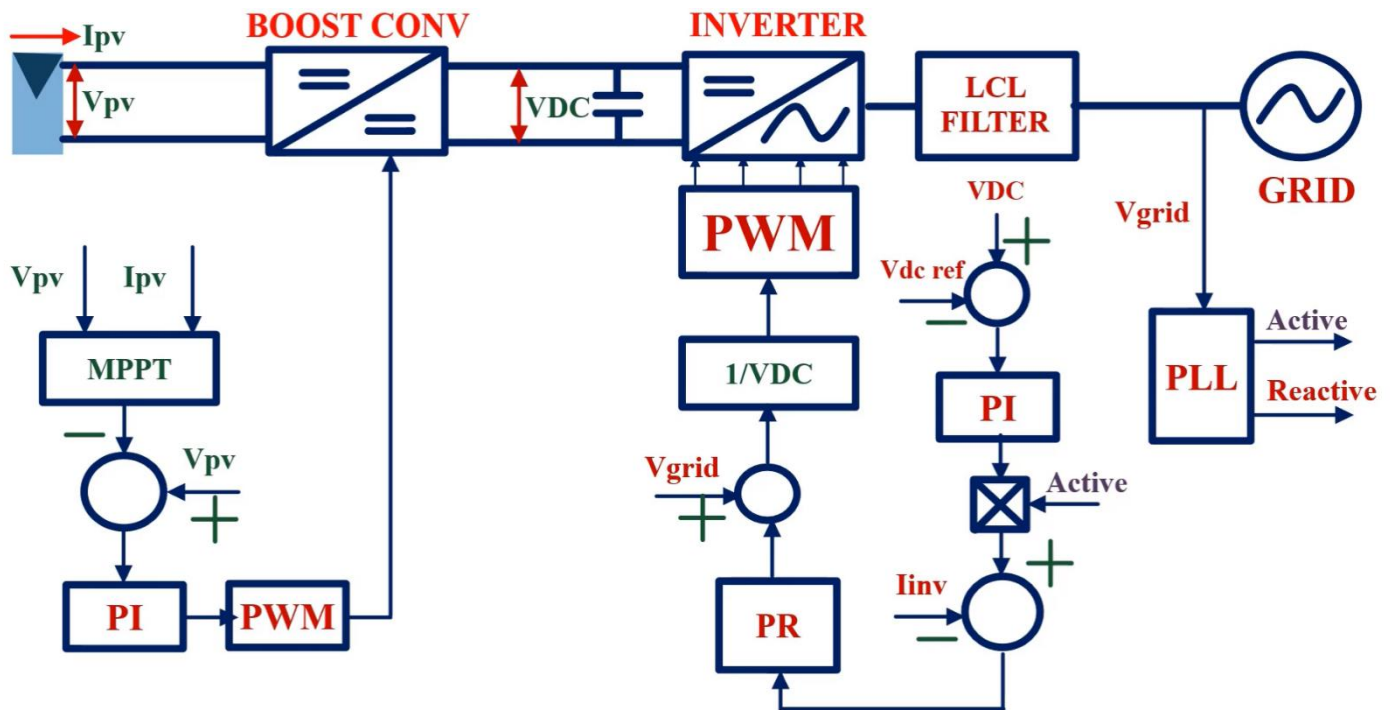


Figure 22: A more robust and detailed version of the whole system.

Below an attempt was made to connect the whole system of PV, Boost converter and single-phase inverter:

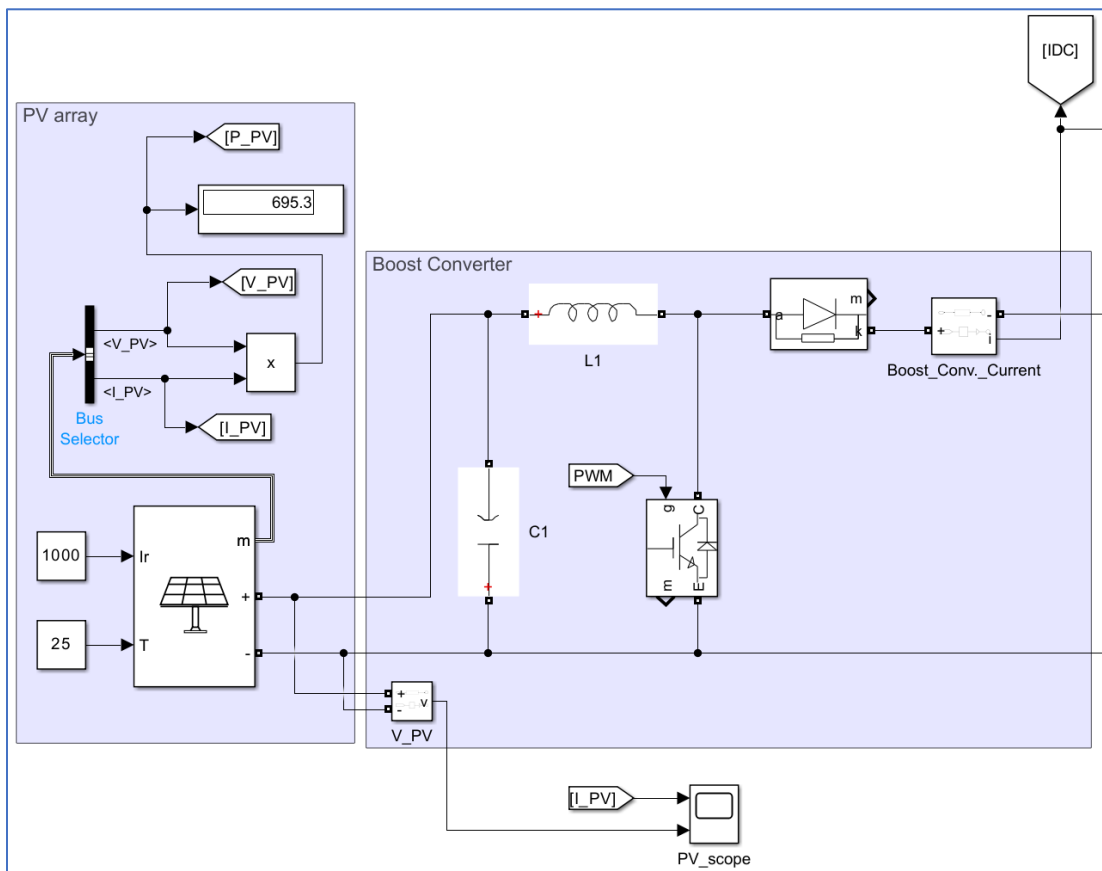


Figure 23: The PV sub-system block and Boost Converter block.

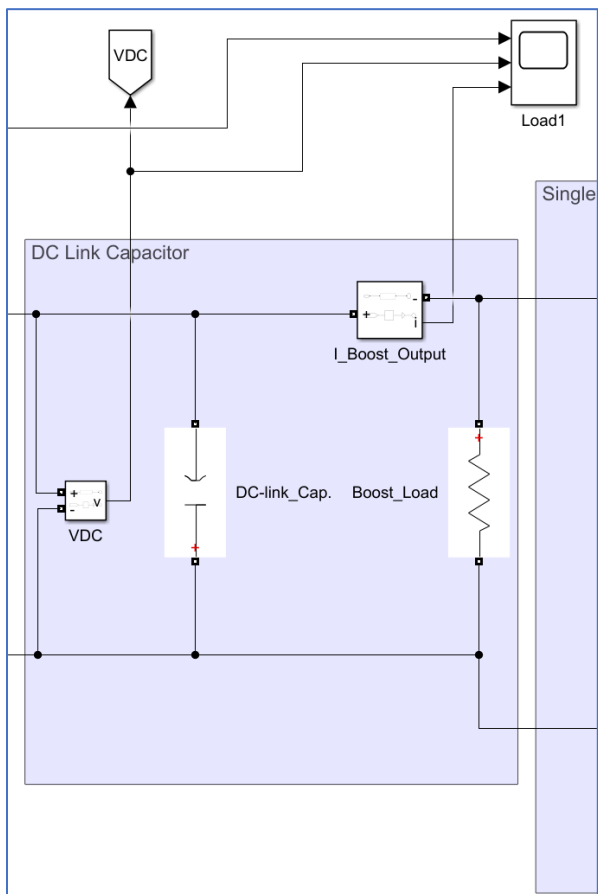


Figure 24: The DC link capacitor connected at the output of the Boost converter.

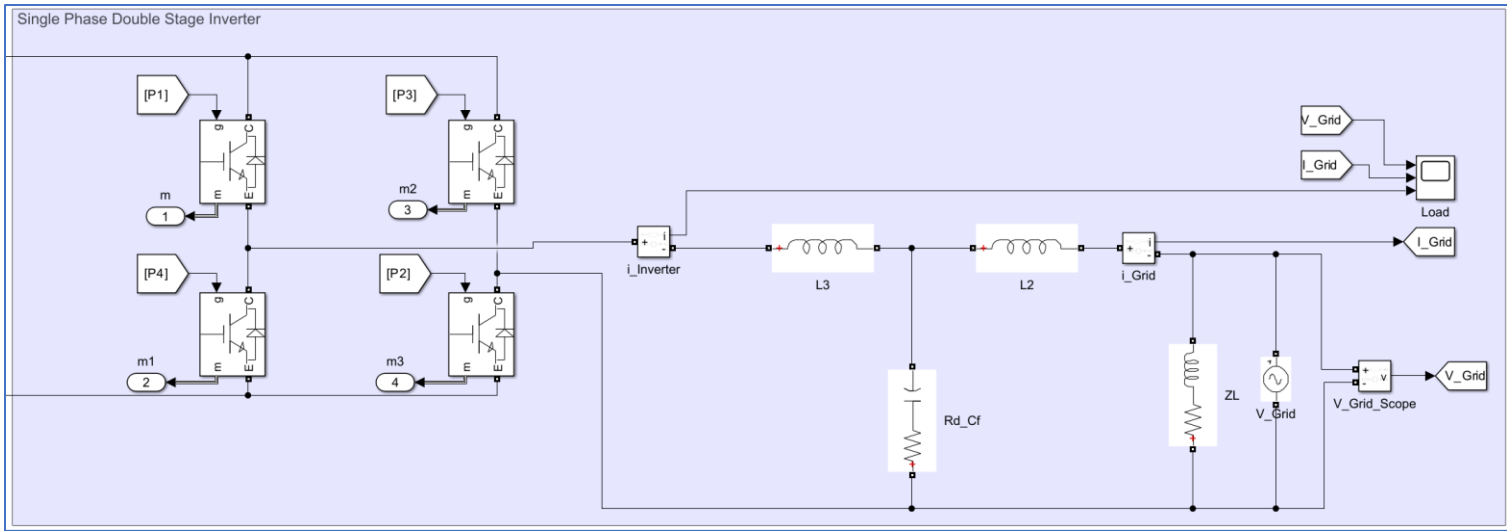


Figure 25: The completed Single Phase double stage inverter connected to the Dc-Link capacitor

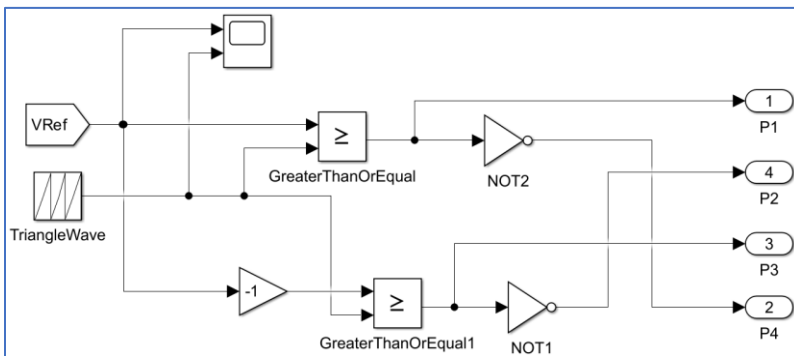


Figure 26: A completed version of Figure 16, Controller logic for the IGBT switches. Notice the Vref. How the Vref signal has been generated will be shown later.

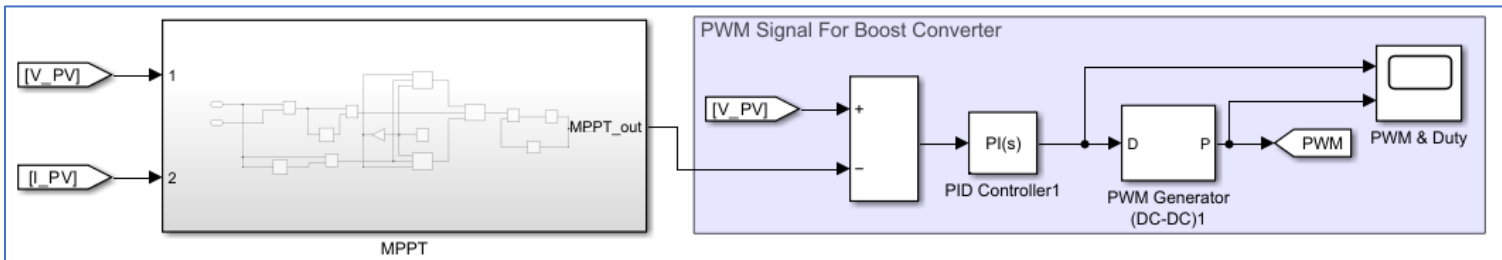


Figure 27: The MPPT controller block and connected to it is the PWM signal generator for boost converter IGBT switch.

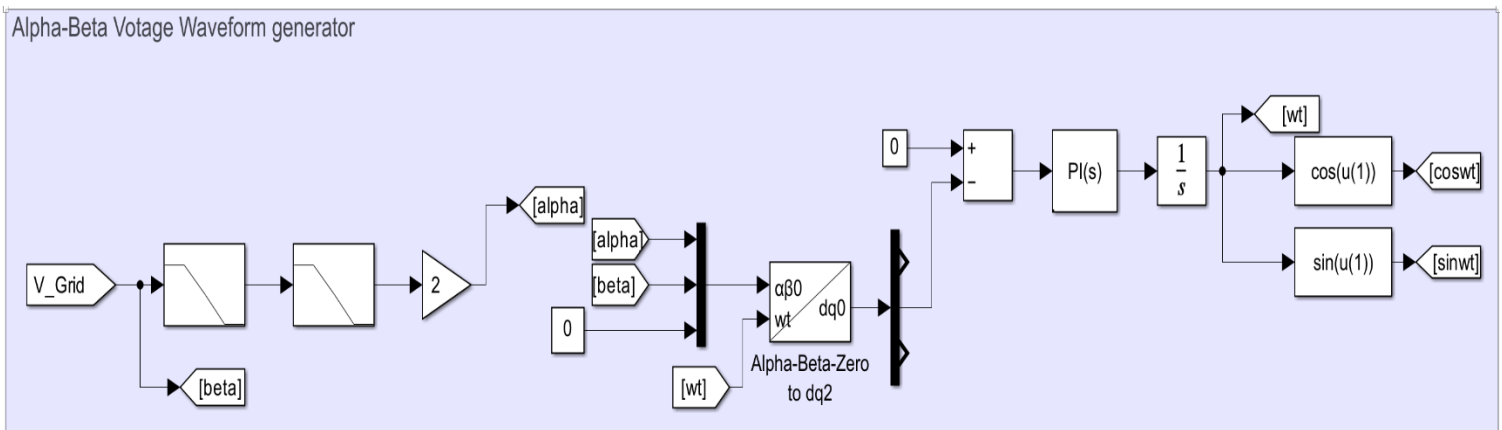


Figure 28: β = Grid voltage and α = 90 degree shifted version of α (PLL sub-system). $\cos wt$ = current reference for active power and $\sin wt$ = current reference for reactive power.

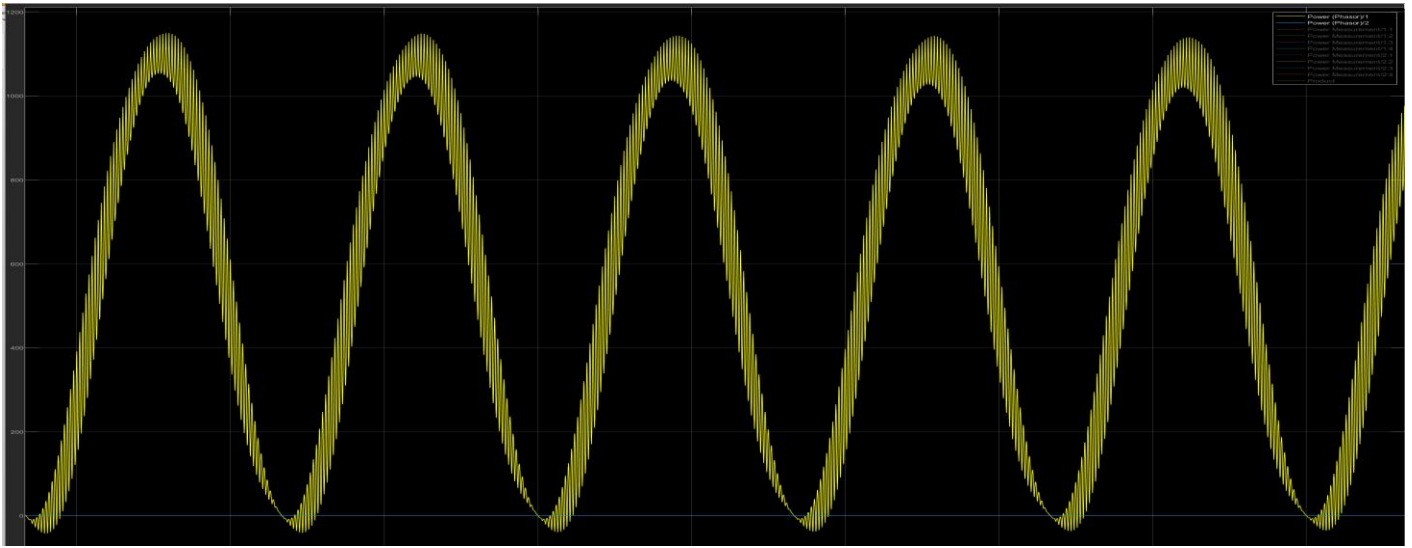


Figure 31: sending around 1000 Watts when irradiance is 800 W/m² and Grid Voltage and current are in phase.

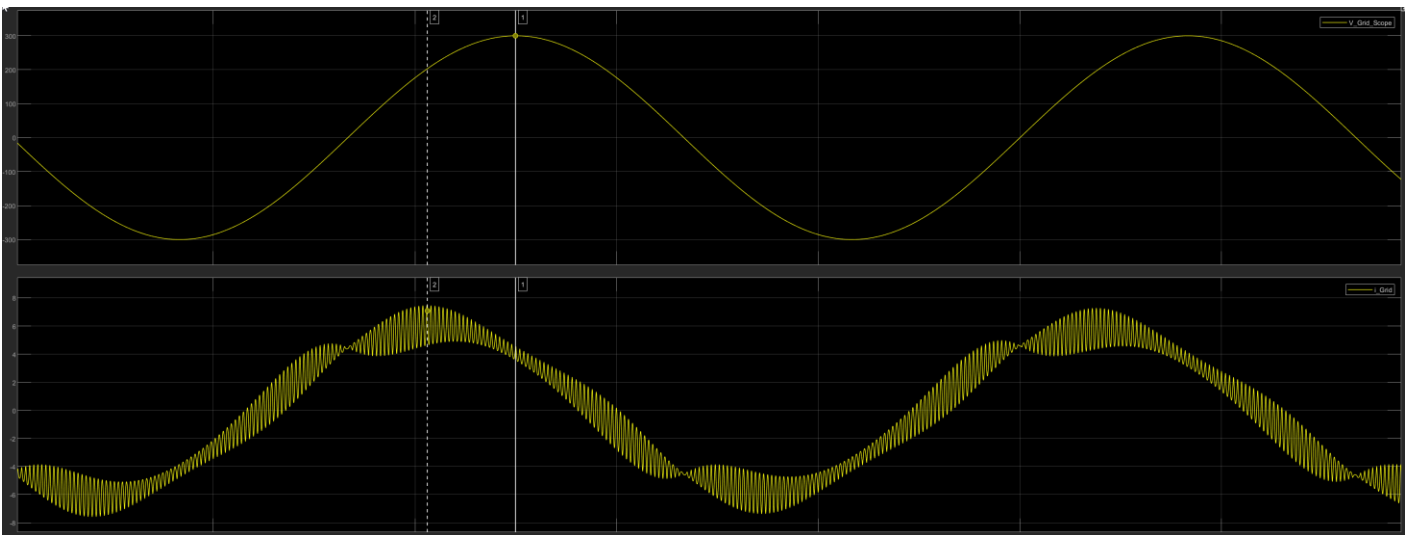


Figure 32: Grid Voltage and current are out of Phase. Sending Re-Active power.

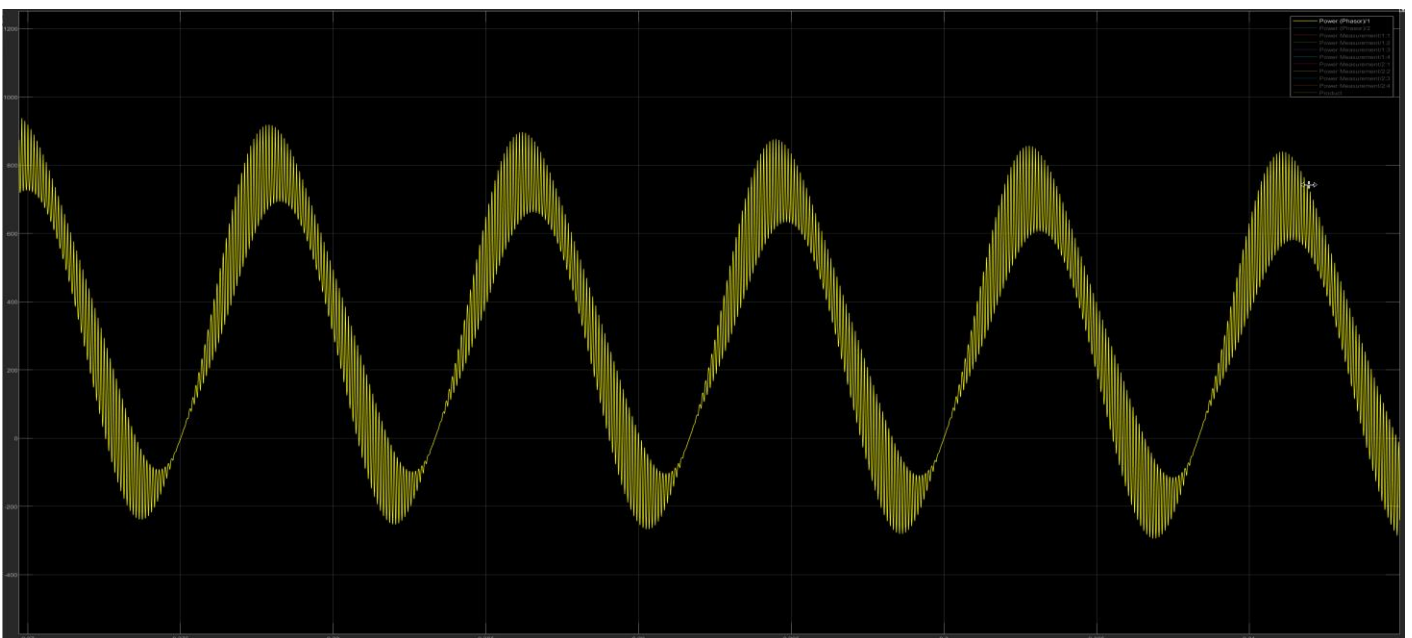


Figure 33: sending around 1000 VAR when irradiance is 500 W/m² and Grid Voltage and current are out of phase.

Some concluding remarks, this is a first pass on a complicated electrical system as a young engineer. There are lots of limitations and improvements can be made by removing those. A grid synchronization subsystem has not been used. That's also an essential part of this kind of system. But, overall, this was a good learning experience. The next steps should be:

- to try to smooth out the output signals better.
- Tuning the PI controller K_p and K_i parameters.
- Adding grid synchronization.
- Modify the PLL to make it easier to understand how to push active and re-active power better.

REFERENCE:

1. Various YOUTUBE videos were used to plan and understand the system.
2. Lecture notes - Purdue University ECE 61600.
3. Project simulation file:
4. [Project-2 Files](#) - Password: 0009988